NIHVA

We Make Possible

# Automation Solutions













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# AUTOMATION
# SOLUTIONS



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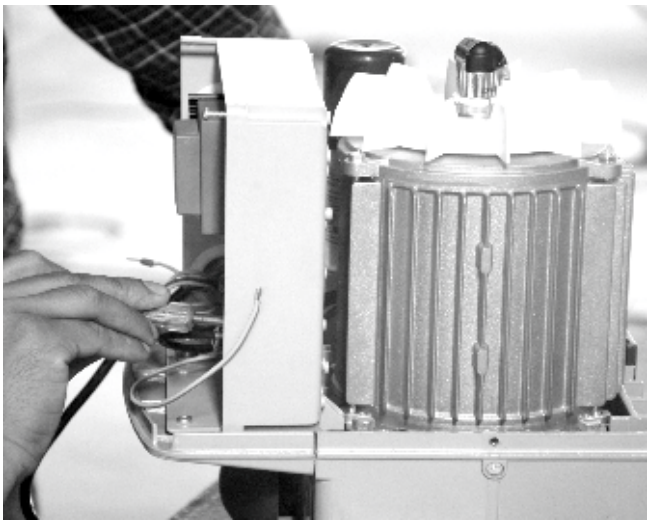
# Special Purpose Machine (SPM)

## Technical Description

- A machine for special applications like ultrasonic welding, shearing, profile punching, Tightening, special drilling, threading, pressing where normal manufacturing process can't be done.
- Widely used in all industries where cycle time reduction is expected with high productivity

## Specification

- Application of product by customer
- Size of part for special purpose application
- Space available for particular purpose
- Height of particular part / access to personnel
- Weight of part
- Type of system to be provided Pneumatic / Hydraulic / Hydropneumatics
- Air pressure consideration
- Power consumption





# Engineering Services

## Assembly / Machining / Hydraulics

## Continuous Fabrication

For successful product we need a team of experts having good workmanship with prior experience. We have a team of such experts & we provide solution to your engineering services, mechanical assembly, fabrication work, hydraulic work.






# Wireless Automation (M2M)

Whether you manufacture for business or consumer, everyone expect highest quality & on time delivery. Many companies implement manual processes to improve or maintain product / process & quality, yet even the most stringent manual processes have major flaw which is human error, human dependability & skill

Automating manufacturing and production processes that removes the inappropriate error and replaces it with standardized design of the product. NIHVA helps in bringing improvement of product quality & process operations.

Sensing systems are used for quality control activities by bringing the product range within acceptable tolerances on the basis of pre-programmed parameters. Products that do not pass QA (Quality Analysis) during the inspection process are rejected or it is directed to waste bin. Process automation system give real-time feedback on product as it moves through assembly line in a plant, and can provide reports on pass / fail rates of product that has passed through quality control. These batches inconsistencies can be tracked and improvement action plan can be initiated to reduce error.

**Real-Time monitor and control solutions**

- Automated quality vision inspection system – QVS
- Production monitoring – scheduling and reporting system

**Monitor & Control**

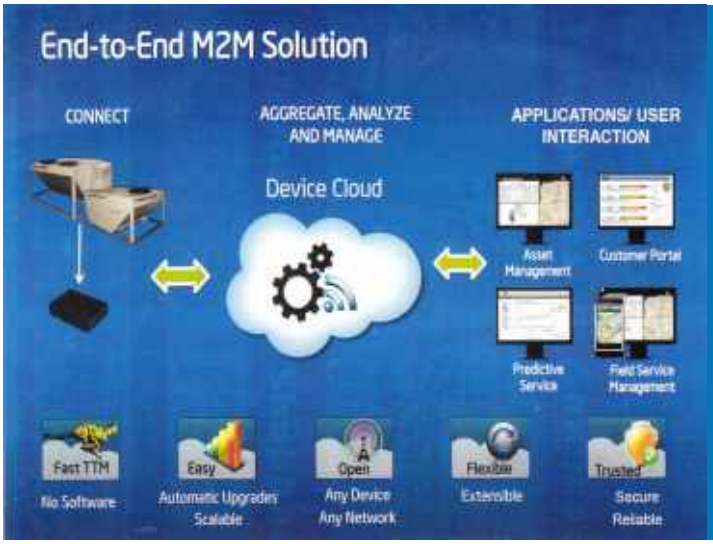
Man | Machine | Material | Movement | Money.




Quality vision system has become a key technology in the area of manufacturing and quality control due to the increasing quality demands of manufacturers and customers. Quality vision system utilizes industrial image processing through the use of cameras mounted over production lines and cells in order to visually inspect products in real time without operator intervention. Nihva Quality vision system is primarily focused on computer vision in the context of industrial manufacturing process, be it in the inspection process itself (e.g. checking a measurement or Identifying a character string is printed correctly) or through some other responsive input needed for control (e.g. robot control or type verification).

Our quality vision system consists of a number of Instrumentation equipment's and cameras all capturing, individually with a control system related to some pre-determined tolerance or requirements. Quality vision system (or automated instrumentation) can be applied in almost any industry. Our modern vision system can be applied across a sectors, diverse range of industry sectors, these industries include:

- **Automotive**
- **Mechanical / MFG**
- **Semi – Conductor**
- **Electronics**
- **Medical Devices / Pharmaceutical**




**Improve your bottom line!**

- In this economy, companies need to find a great balance between lowering costs and maintaining the same level of quality of product and efficiency of the system – both yielding better ROI. The critical factors for your business will now be in your control.
- Our M2M solutions allow for companies to catch issues earlier then before, make changes and decrease the amount of downtime on the floor…. Sounds like smart business!



# Material Handling Equipments



## Telescopic Conveyor

Capacity - 40 to 75kg/m

A material handling equipment for loading & unloading of individual product which are shipped in trailer. 20ft or 40 ft like Cartons tea bags, sacks, etc. with economical & eco hands-free operations loading and unloading vehicle becomes easy with conveyor which consumes min. manpower.



## Goods Lift

Capacity - 250 to 2000kg

A product which reduces human efforts of lifting material & delivering to different level & floors. A simple platform which can be controlled with simple operating panel & ensures load capacity to 3 ton to 6 ton



## Scissor Lift

Capacity - 500 to 12000kg

A working platform to reached at a certain height for maintenance or any other work. A platform on which we can carry a heavy material along with. Which is controlled by pendant & operating panel.



## Manipulator

Capacity - 50 to 200kg

A manually controlled robo-arm to handle material with different axis, by hydraulic or pneumatic supply.



## Maintenance Work Platform

Capacity - 50 to 300kg

Economics & fully manual mechanical platform to work at several heights. It is a best option to replace scaffolding to easy & quick assembly solution.

# Robotics Applications

We design and build turn-key solutions using multi-axis robot arms, custom end-of-arm tooling, conveyance systems etc. We strive to integrate the best design, and technology, all the time maintaining a customer focus to produce solutions for your specific production environment. Our partnership with the key Robotic Arm Manufacturers enables us to provide you with the right automated solutions for your application. Our Engineers focus on the customer's process and strive to understand their needs in order to design and build innovative manufacturing solutions. Our engineers can provide more information about our services and custom robotic technologies. Contact us today for an engineering review of your needs.



# Automatic Sliding Roof



Automatic Sliding Roof is a new innovation by NIHVA for bungalows, hotels, hospital, villas, apartments, commercial complex, offices, where we can add more glory by allowing partial sunlight to enter in premises at day and enable sky view at nights. Roofs are automatically operated in many ways like folding, sliding, telescopic, single sided, double sided, vertical lift, inclined lift etc.

Made of aluminium profiles, water-resistant, temperature resistant, wind-resistant, sealed, one touch control, available in various sizes and glass specifications.

Equipped with advanced electronic control system with low operating currents and a dedicated building management system (BMS) input. Many other automation inputs possible with prior consultation.





INNOVATIONS

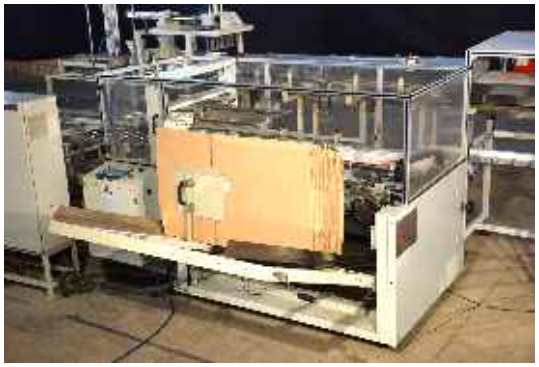
# Secondary Packaging Machines

- Secondary packaging includes cardboard cartons, cardboard boxes, cardboard/plastic crates, trays, paperboard cartons, shrink wrapped bundles, trayed PET bottles held with shrink wrap, taped or glued cardboard cartons containing cans of food, Pouches of food, Consumable material.
- Loading of cans, pouches in cartons known as secondary packaging.
- Secondary packaging is an excellent way to reinforce your brand and enhance product visibility in a crowded marketplace.







**Palletizer :**



A palletizer is a machine which provides automatic means for stacking cases (cartons) of goods or products onto a pallet. Manually placing boxes on pallets can be time consuming and expensive; it can also put unusual stress on workers.

There are specific types of palletizers.

1) In row-forming palletizing applications loads are arranged on a row forming area and then moved onto a different area where layer forming takes place. This process repeats until a full layer of goods and products are configured to be placed on a pallet.

2) The in-line palletizer when higher speeds were needed for palletizing. This palletizer type utilizes a continuous motion flow divider that guides the goods into the desired area on the layer forming platform.



**Pallet Dispenser:**

Pallet Dispenser is a machine which Dispenses the magazine stack of pallet one by one on conveyor system of palletizer. It has a capacity to hold load of stacked pallet. Stack of pallet can be designed as per requirement.

The pallet dispenser is operated by a pneu matic scissor-lift mechanism. magazine stack of pallet is placed on Scissor lift , once the scissor lift reaches to Dispenser arm position it hold stack of pallet with single pallet on scissor lift which drops on conveyor system for further palletization

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- Dock Houses

## Automatic Sliding Roofs

## Water Management & Environment Solution

## Material Handling Equipments

## Automation Solutions

## NIHVA Service BACK-UP (24/7)

# Opening doors to business and life



We Make Possible







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## International Business Partners

ALBANY ASSA ABLOY | MEGADOOR ASSA ABLOY | Crawford ASSA ABLOY | besam ASSA ABLOY | ACO

## Networks

| WEST | NORTH | SOUTH | EAST |
|---|---|---|---|
| Pune Mumbai Nashik Nagpur Indore Ahmedabad Surat Vadodara | Delhi Chandigarh | Chennai Hyderabad Bengaluru | Kolkata Bhubaneshwar |